

ANGULAR DEVELOPER PRACTICAL ASSESSMENT

Task Focus: Logical Ability Test (Weather Application)

UI Framework: Tailwind CSS

1. CORE OBJECTIVE

Build a highly responsive Weather Dashboard in Angular that connects to the OpenWeatherMap API (<https://openweathermap.org/>) to display real-time weather analytics and multi-day forecasts.

2. CORE REQUIREMENTS

- **Current Weather Metrics:** Display the precise current temperature in Celsius. The application must target user location permissions or default to 'Bengaluru'. Provide an interactive search input to dynamically alter the active city location.
- **Extended Forecast Display:** Integrate the forecast5 or forecast16 API endpoints. Render an updated dashboard interface highlighting a complete 7-day extended timeline (including today) along with mathematical averages calculated for the upcoming 4 days.
- **Network Mutation States:** Incorporate responsive async placeholder blocks. Build a preload/progress bar with a solid blue background while pipeline requests are pending. On request failures, intercept gracefully with a yellow retry interface trigger.

3. TECHNICAL CONSTRAINTS & EVALUATION FOCUS

1. **RxJS Stream Architecture:** Mandatory deployment of RxJS operator strategies (e.g., debounceTime) on the search input element to cleanly choke fast input triggers and stream concurrent asynchronous executions.
2. **State & Security Infrastructure:** Protect application entry blocks—only authenticated, logged-in user identities can bypass access guards. Maintain robust UI data transforms and flexible state layers (via custom services or clean reactive data stores).
3. **Responsive Interface Layer:** Fully structure the design footprints using Tailwind CSS rules. Components must naturally adapt across mobile screens and desktop configurations with absolute layout consistency.

💡 **Engineering Guideline:** Code quality must emphasize clear encapsulation, performance consideration, and structural modularity. Write granular unit test specs targeting your critical state services and view components.

4. GIT CAPABILITIES & SUBMISSION RULES

The deployment and version lifecycle will form a major part of the overall score. Ensure alignment with the following standards:

- Deploy the live builds to **GitHub Pages** ensuring all endpoints run smoothly in production environments.
- Maintain structural clarity inside the repository through explicit branching strategies.
- Ensure atomic commits with concise, expressive messaging documenting incremental history changes.

Forward the finalized GitHub Pages live link via email notification once complete.